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Review

Smart City Applications to Promote Citizen Participation in City Management and Governance: A Systematic Review

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Abstract: This systematic review aimed to provide a comprehensive view of (1) the purposes of research studies using smart city infrastructures to promote citizen participation in the cities' management and governance, (2) the characteristics of the proposed solutions in terms of data sources, data quality, and data security and privacy mechanisms, as well, as strategies to incentivize citizen participation, and (3) the development stages of the applications being reported. An electronic search was conducted combining relevant databases and keywords, and 76 studies were included after a selection process. The results show a current interest in developing applications to promote citizen participation to identify urban problems and contribute to decision-making processes. Most of the included studies considered citizens as agents able to report issues (e.g., issues related to the maintenance of urban infrastructures or the mobility in urban spaces), monitor certain environmental parameters (e.g., air or acoustic pollution), and share opinions (e.g., opinions about the performance of local authorities) to support city management. Moreover, a minority of the included studies developed collaborative applications to involve citizens in decision-making processes in urban planning, the selection of development projects, and deepening democratic values. It is possible to conclude about the existence of significant research related to the topic of this systematic review, but also about the need to deepen mechanisms to guarantee data quality and data security and privacy, to develop strategies to incentivize citizen participation, and to implement robust experimental set-ups to evaluate the impact of the developed applications in daily contexts.

Keywords: smart cities; citizen participation; crowdsourcing; crowdsensing; systematic review

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1. Introduction

Citizens represent the lifeblood of a city and are inextricably linked to its continued existence and prosperity. Therefore, it is imperative for the government of a city to care for its citizens and to pay careful attention to their needs [1]. Moreover, citizens must be active in their cities' management and governance, since they are aware of the problems of the communities where they live and work and can evaluate the actions of city authorities [2,3].

However, despite its importance, citizen participation in the management and governance of their cities is not a simple process, even when city authorities value citizen opinions [4]. To optimize this process and face diminishing public trust due to scandals, corruption, worsening of the economic situation and inequalities [5,6], city authorities are changing and updating government mechanisms to increase citizen participation [7–9]. This has naturally led to the formulation of concepts such as e-government and e-governance, where information technologies are used to deliver services and facilitate communication